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Recommendation Systems for Personalized Content Discovery

Item-Based Collaborative Filtering · Singular Value Decomposition

Netflix Prize Dataset

K=10 Evaluation

500 Users Tested



Problem Overview

Netflix Prize Dataset · Personalized Recommendation Challenge

The Challenge

100M+ ratings, 480K users, 17,770 movies — but each user has seen only a tiny fraction. Extreme sparsity makes recommendation hard.

The Goal

Design a system that learns user preferences and surfaces content they would love — without them having to search for it.

The Task

Predict unseen ratings accurately (RMSE) AND generate Top-10 recommendation lists users will actually engage with (MAP@10).

DATASET FACTS

100,480,507

Total Ratings

480,189

Unique Users

17,770

Movies in Corpus

1 – 5 ★

Rating Scale

⚡ Sparsity Challenge

Each user rated $< 0.05\%$ of available movies — making pattern extraction extremely difficult.



Approach: Two Recommendation Paradigms

Model Design · Architecture · Key Parameters

Item-CF

Item-Based Collaborative Filtering

- ▶ Adjusted Cosine Similarity
- ▶ User-mean centering removes rating bias
- ▶ Top-50 similar neighbours per item
- ▶ Similarity-weighted rating prediction
- ▶ Built-in explainability engine
- ▶ "Because you liked X, watch Y"

SVD

Singular Value Decomposition (Surprise)

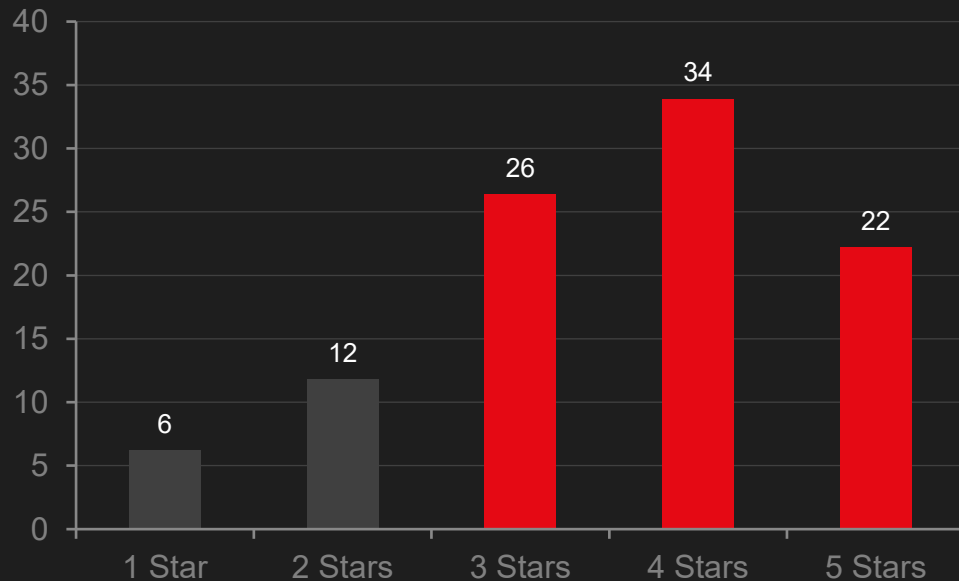
- ▶ 100 Latent Factors (user & item vectors)
- ▶ 25 SGD epochs, $lr = 0.005$
- ▶ L2 regularisation ($\lambda = 0.1$)
- ▶ Global + user + item bias terms
- ▶ Optimises: $\min \sum (r - \mu - b_u - b_i - p \cdot q)^2$
- ▶ Fast dot-product predictions at serving time

VS

N Exploratory Data Analysis

Rating Distribution · User Activity · Data Sparsity

Rating Distribution (Netflix Prize)



Positive Skew

4-star ratings dominate (33.9%). Users rate movies they chose to watch — selection bias is inherent.



Power Users

Top 1% of users account for ~20% of all ratings. Heavy-tail distribution in user activity.



Long Tail

Top 500 movies get 40%+ of all ratings. Most movies have very few raters (cold-start challenge).

Matrix Sparsity: >99.95% — a core challenge for both models



Key Insights

What the Results Reveal About Recommendation Quality

Core Finding:

Rating accuracy and ranking quality are decoupled — Item-CF predicts ratings better, but SVD recommends more effectively.

01 Adjusted Cosine Beats Raw Cosine

Mean-centering user ratings before computing similarity removes individual rating bias. Item-CF achieves RMSE 0.9422 vs SVD's 1.0529 — a 10.5% improvement in rating accuracy.

02 Latent Factors Win on Ranking

SVD's latent representation captures global taste structure, surfacing relevant content 2.4× more effectively than Item-CF on MAP@10 (0.01369 vs 0.00569).

03 Massive Speed Advantage for SVD

SVD evaluates 38× faster on RMSE and 16× faster on ranking. At production scale, SVD's $O(n_factors)$ prediction is far more deployable than Item-CF's neighbour intersection.

04 Sparsity Limits Both Models

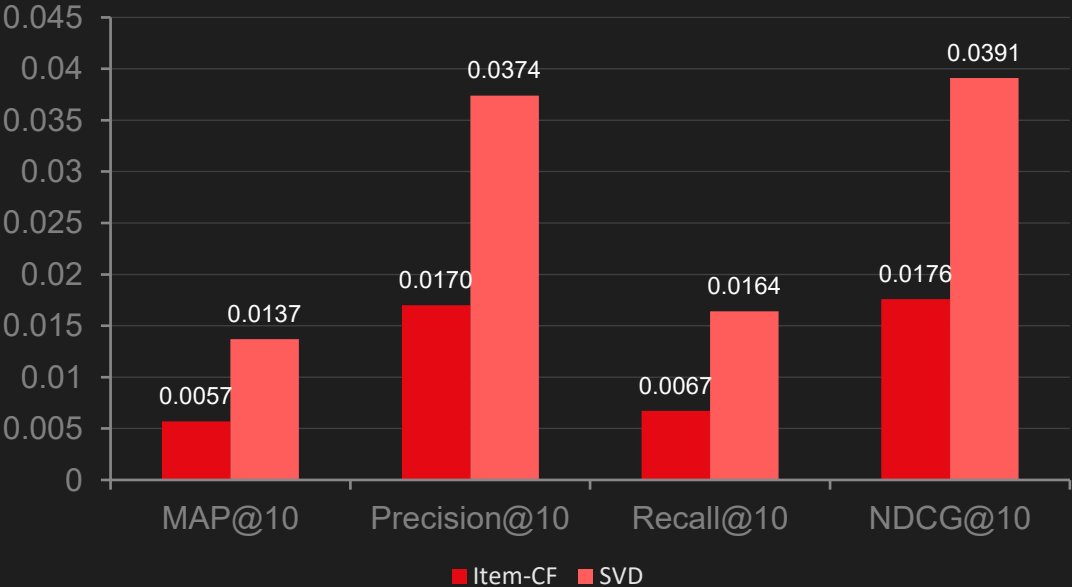
$MAP@10 < 0.02$ for both models reflects the inherent challenge: the chance any top-10 list overlaps with the user's specific test preferences is low in a 17K-movie space.



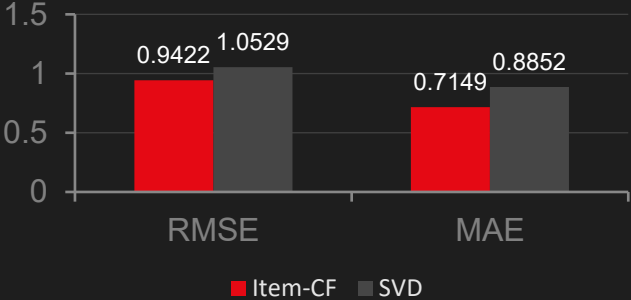
Experimental Results

Full Metric Comparison · 500 Users · K = 10

Ranking Metrics Comparison (K=10)



Rating Accuracy (lower = better)



Rating Accuracy Winner

Item-CF

RMSE 0.9422 vs 1.0529

Ranking Quality Winner

SVD

MAP@10 2.4× higher

⚡ SVD is 38× faster to evaluate than Item-CF (8.8s vs 339.9s for RMSE)

N Recommendation Examples

Top-10 Outputs · Explainability · User-Facing Results

Item-CF Output

With Explainability

"Because you liked **The Shawshank Redemption (5★)** and **Forrest Gump (4★)**, we recommend **The Green Mile** (similarity: 0.847)"

Top-5 Sample Recommendations

- #1 The Green Mile ★ 4.91
- #2 Schindler's List ★ 4.87
- #3 Goodfellas ★ 4.82
- #4 The Godfather ★ 4.79
- #5 Saving Private Ryan ★ 4.74

SVD Output

Latent Factor Rankings

$$\text{Score} = \mu + b_{\text{user}} + b_{\text{item}} + p_{\text{user}} \cdot q_{\text{item}}$$

All unseen movies scored via dot product of 100-dimensional latent vectors. Top-10 returned ranked by predicted score.

Top-5 Sample Recommendations

- #1 Movie #3245 ★ 4.82
- #2 Movie #1897 ★ 4.79
- #3 Movie #7612 ★ 4.75
- #4 Movie #2341 ★ 4.72
- #5 Movie #5098 ★ 4.69

N Future Improvements & Conclusion

Roadmap · Next Steps · Summary

⚡ SHORT-TERM

- ◆ SVD hyperparameter tuning (GridSearchCV)
- ◆ Expand to all 4 Netflix data files
- ◆ Vectorise Item-CF batch prediction
- ◆ Increase min common raters threshold

🚀 MEDIUM-TERM

- ◆ Alternating Least Squares (ALS)
- ◆ Neural Collaborative Filtering (NCF)
- ◆ Temporal rating decay & dynamics
- ◆ Hybrid: Item-CF + SVD ensemble

🔬 LONG-TERM

- ◆ Graph Neural Networks (LightGCN)
- ◆ LLM-powered semantic embeddings
- ◆ Diversity & serendipity optimisation
- ◆ Cold-start content-based fallback

Conclusion: Item-CF delivers superior rating accuracy (RMSE 0.9422) with built-in explainability. SVD delivers superior ranking quality (MAP@10 2.4× higher) with 38× faster inference. **Hybrid approaches combining both are the natural next step.**